LESSON 11 · MISSION GUIDE

Show Gerald

Charts people can read, and Excel files people can open.

From **Gerald Brightwell, Managing Director**
To **Jen**
Subject **Re: Re: Re: the numbers (sent Friday, 16:55)**

Jen, Priya showed me what you've been doing with the "pandas". Very impressive. I understood almost half of it.

The board meets in two weeks. The board does not read tables. The board reads *pictures*, ideally blue ones, and then asks for "the Excel" to fiddle with on the train. Dave did this by hand over a weekend, which is why Dave now lives on a boat. Can your Python draw the pictures and make the Excel? With bold headings? Have a lovely weekend. — G.

What you'll be able to do after today

- [ ] Draw a **bar chart** and a **histogram** from a DataFrame with matplotlib, and save them as PNG files
- [ ] Put two series **side by side** (men vs women per department) and label everything properly
- [ ] Write a DataFrame to an **Excel workbook**, and several tables to several **sheets** of one workbook
- [ ] Format a workbook with openpyxl: bold coloured headers, `£#,##0.00` number formats, column widths, frozen panes
- [ ] Drop a chart image into a worksheet

Why this matters at work

- **The last mile is the one people see.** Nobody on the board will run your script. They'll open a PNG in an email and an xlsx on the train. If that looks right, the whole project looks right.
- **Formatting by code is formatting forever.** Set the header style once in Python and every month's report has it. No more "can you make it look like last month's".

Four big ideas

1. A chart is a file	In a script there is no window. `plt.savefig("x.png")` writes the picture; you open it like any file. `matplotlib.use("Agg")` at the top tells matplotlib not to look for a screen.
2. Build, then decorate	Order of a chart: `plt.figure()` → draw (`bar`, `hist`) → title, labels, legend → `tight_layout()` → `savefig()`. Same every time.
3. pandas writes Excel	`df.to_excel(path, sheet_name=...)` for one sheet; `with pd.ExcelWriter(path) as writer:` for several. Needs openpyxl installed, which is why it's in `requirements.txt`.
4. openpyxl edits the workbook	pandas writes values. For looks, re-open with `load_workbook`, walk the cells, set `font`, `fill`, `number_format`, widths, then `save`.

Things that will bite you

- Calling `plt.show()` in a script: it either hangs or does nothing. Save instead.
- Drawing a second chart without `plt.figure()`: both charts land on the same picture.
- Excel number formats are text, with a pound sign: `"£#,##0.00"`. Exactly the string you'd type in Format Cells.
- openpyxl colours have no `#`: `"1F3A5F"`. matplotlib colours do: `"#1f3a5f"`. Yes, really.

How today works

15 MIN
Lecture
Lars draws two charts and writes one workbook, live

60 MIN
Exercise sheet
Seven tickets: three charts, four workbooks

15 MIN
Dave's Leftovers #11
The board pack in one click. Gerald clicked.

The inbox

#	FROM	TICKET	YOU'LL USE
1	Gerald	Your first chart	`plt.bar savefig`
2	Siobhan	Where do salaries sit?	`plt.hist axvline legend`
3	Priya	Side by side	`pivot_table np.arange` grouped bars
4	Mo	Back to Excel	`to_excel`
5	Priya	One workbook, several sheets	`pd.ExcelWriter`
6	Gerald	Make it pretty	`load_workbook Font PatternFill number_format`
7 ⭐	Gerald	A chart inside the workbook	`barh add_image`

You're done when...

- [ ] `my_work/output/` holds four PNGs and four workbooks, and you've opened every one of them
- [ ] Every script prints what the sheet says, and Dave's board pack names the right top three
- [ ] Everything is pushed to `lesson-11`, with your favourite chart attached to the pull request



Next lesson: Retire MACRO-9000. The finale. Two sessions, one report: the Brightwell gender pay gap, by the book, with a chart, a workbook, and one sentence for the board. Everything you've learned, in one project folder.